

Vibhore Mathur

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EDUCATION

B.Tech (Computer Science and Engineering)

Swami Keshvanand Institute of Technology, Jaipur

2022–2026

CGPA: 8.12/10

Senior Secondary (Class XII)

Ryan International School, CBSE

2022

82.8%

Secondary (Class X)

Ryan International School, CBSE

2021

92.2%

EXPERIENCE

Outbox Labs | Software Developer | Bengaluru, Karnataka

October 2025 – Present

- Built features across the **Zapmail** platform, including deliverability, domains, workspaces, automation, and platform reliability.
- Developed a domain health engine for automated blacklist monitoring and DNS validation, improving email deliverability reliability.
- Engineered a centralized **BullMQ** + **Redis** queue system with retry/backoff handling for API rate limits, reducing job failures by **40%**.
- Built a high-throughput webhook delivery system processing **15,000+ events/day** with **HMAC verification**, idempotency, and configurable retries.
- Led engineering for a major product redesign, delivering key user flows that improved onboarding completion and platform retention.

SoftSensor.AI | MERN Developer Intern | Jaipur, Rajasthan

May 2025 – August 2025

- Built a centralized **MQTT alert system** delivering **1,000+ cross-app notifications/week** and collaborated with cross-functional teams in an **Agile** environment to develop scalable full-stack modules.
- Designed a unified **DTO architecture** across **30+ backend routes**, eliminating inconsistent API responses, reducing client-side defects, and improving long-term maintainability.

PROJECTS

• CSVQL – Custom SQL Interpreter with Terminal UI



Tools: Python, Textual, Regex, Dataclasses

- Built a **5-stage SQL compiler pipeline** (Lexer → Parser → AST → Planner → Executor) from scratch in Python, executing SQL queries directly against **CSV files** with zero external dependencies.
- Implemented a **recursive-descent parser** supporting the full SQL grammar — **JOINS** (INNER, LEFT, RIGHT, FULL OUTER), subqueries, CASE/WHEN, GROUP BY, HAVING, and DML/DDL — backed by a **typed AST** of 15+ node types using **dataclasses**.
- Developed a **Textual-based TUI** featuring a live schema browser, syntax-highlighted SQL editor, **real-time prefix autocomplete**, and a query status bar reporting execution time in milliseconds.

• Stratifyy – Stock Backtester

[Live Demo](#) |

Tools: React (TypeScript), Node.js, C++, Express.js, PostgreSQL, Socket.io, Recharts, JWT

- Implemented multiple trading strategies including **Bollinger Bands**, **Mean Reversion**, and **Momentum**, leveraging a **C++ engine** for **high-performance computations** and enabling flexible, configurable backtests.
- Built **interactive dashboards** with equity curve visualizations, **real-time trade updates via webhooks**, and drawdown analysis for key performance metrics.

• Email Sync RAG Pipeline

[Live Demo](#) |

Tools: Node.js (TypeScript), GraphQL, MongoDB, Elasticsearch, Qdrant, Docker, React (TypeScript), IMAP

- Built a full-stack **RAG pipeline** for real-time email classification and AI-assisted replies, integrating **OpenAI API** with automated Slack notifications.
- Engineered IMAP synchronization, email parsing, and embedding storage using **Qdrant**, with indexing and retrieval optimized via **Elasticsearch**.

SKILLS

- **Languages & Core Technologies:** JavaScript, TypeScript, Python, C++, React.js, Node.js, Express.js, GraphQL, Socket.IO
- **Infrastructure, Databases & Tooling:** PostgreSQL, MongoDB, Redis, AWS, Docker, BullMQ, Qdrant, Playwright, Git, OOPS